

Random Punch Out

Time limit per test: 1 second
memory limit per test: 256 megabytes
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In the classic boxing Game "Punch-Out!!", for the NES, the main character, Little Mac, fights many different boxers in sequence, in order to get to the final adversary, Mike Tyson. However, in a corrupted version of the game's cartridge, two modifications happened inexplicably:

- Now, Little Mac does not regenerate any health between fights.
- Also, instead of starting the sequence on the first fighter, the game now starts on a randomly selected fighter out of all fighters.



Since you are an expert player, you know how to play every level optimally, which means you know how to defeat the i -th fighter losing the minimal amount of health, a_i . More formally, if you are on level i and Little Mac's health is h , you can defeat the i -th fighter if and only if $h \geq a_i$, and in that case your new health value becomes $h - a_i$ and you move on to level $i+1$. Otherwise, if $h < a_i$, you run out of health and the game ends.

Let's define the score of a run as the number of fighters defeated before Little Mac runs out of health. You want to know: given a initial health h (fixed for all runs), and knowing the health loss a_i for each fight i , what would be the maximum score you can get?

Note: the maximum score you need to find will be defined as the maximum of the scores obtained starting at the first, second, ..., n -th fighters.

Input

In the first line, there will be two integers n, h , with $1 \leq n \leq 10^5$ being the number of fighters and $0 \leq h \leq 10^{18}$ the initial health value.

In the second line, there will be n non-negative integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$), with a_i being how many health points the i -th fight costs.

Output

Output a single integer: the maximum score possible. Note that this score might be 0.

Examples

Input	Output
7 20 2 6 4 3 6 8 9	4
7 7 2 6 4 3 6 8 9	2
7 1 2 6 4 3 6 8 9	0

Explanation of test case 1:

- Starting at the first fighter, the best possible run defeats fighters 1 through 4 (since $2 + 6 + 4 + 3 \leq 20$, but $2 + 6 + 4 + 3 + 6 > 20$), 4 points. Starting at fighter 2, the best possible run ends on fighter 5, getting 4 points.
- Starting at fighter 3, the best possible run ends at fighter 5, getting 3 points.
- Starting at fighter 4, the best possible run ends at 6, getting 3 points.
- Starting at fighter 5, the best possible run ends at fighter 6, getting 2 points.
- Starting at fighter 6, the best possible run ends at fighter 7, getting 2 points.
- Starting at fighter 7, the best possible run ends at fighter 7 itself.

The answer will then be $\max(4, 4, 3, 3, 2, 1) = 4$.

Explanation of test case 2:

The best run is starting at fighter 3 for 2 points, since $4 + 3 \leq 7$.

Note some runs can earn 0 points (like starting at the last fighter, which can't be defeated with only 7 health points).